

Project 1: Intro

About the data

The dataset contains data from two traffic monitoring points in Frogner, Oslo: one monitoring motor vehicles and one monitoring bicycles. You will also be given weather data from a nearby location.

The data are provided in the original format. The traffic data cover the period from 1 January 2026 to 1 June 2026. Your job is to work as a data science team, finding out how the data can be used, i.e. gaining knowledge and value from data.

Overall aim

The overall aim of the project is modelling of traffic load (number of vehicles per hour) using the two different data sources (traffic monitoring and weather).

You will use multiple linear regression and can apply all variations of this that you have learnt so far (transformations, categorical variables, interactions), or add new elements.

Before you can arrive at a suitable model, you will have to make several **choices**. Some choices are up to your imagination (no right/wrong), but for most problems there are more and less reasonable choices to make.

Grading

In general, we are looking for both **correctness** in use of methods, and your **understanding/reflections** regarding the data, your choices and your results.

AI-use: We expect you to take advantage of all available resources, but the challenge is to use them in a good way.

Documenting progress: One of the hand-ins for the project is a (brief) timeline. At the end of *each* session you will make a short reflective note that by the final delivery will contribute to the timeline.

Part 1: Getting to know the data, data cleaning, and prepare for modelling

Part 1 is about the first steps leading up to modelling, not the modelling itself.

The data sets

The file `frogner_traffic.csv` contains data from both the motor vehicle and bicycle monitoring points. The column `Trafikkmengde` contains counts of vehicles in a 1 hour interval (traffic load), per lane (`Felt`) and in total.

Details regarding the data collection and presentation can be found (in Norwegian) at <https://trafikdata.atlas.vegvesen.no/om-trafikdata>

The data has not been cleaned for you, and you need to e.g. handle missing data and make sure columns of numbers are not represented as characters.

An important **choice** you will have to make is what traffic load to model. Total? Lane one? Bicycles? Light vehicles? Note that there is no right and wrong choice, but some choices might be more suited for a linear regression model than others.

The dataset gives the date and time of the counts, and from this you can **choose** to construct temporal variables (variables associated with time) that enter the model, e.g. weekday or weekend. You can also **choose** to model only certain time periods.

The file [frogner_weather.xlsx](#) gives weather data from a weather station nearby the traffic monitoring point. You need to deal with different time resolution when you combine the two data sources, and what weather variables to enter into your model.

Note that the actual modelling happens in Part 2 of the project, in Part 1 you are getting to know and defining the response variable (traffic load) and the covariates (e.g. day of the week and weather).

Deliverables Part 1

2 reflection sheets (Wednesday and Friday)

1 modelling pipeline (Friday)

Modelling pipeline

1. Modelling Aim. Decide on an aim for your model of traffic: What do you as a group want to strive for? How will you balance interpretability (usually few covariates) and model fit (the models ability to explain variability in the outcome)? Is your study exploratory, or do you have a specific hypothesis that you want to investigate?
2. Visualizations. Present figures of your chosen response (traffic count). The figures can be drafts that you improve later on.
3. Select weather variables, explain your choice and discuss how they might be added to your model (linear effect, transformations, categorical variables, splines?)
4. Create temporal variables based on the date and time of the traffic counts. Which variables did you create, and how will you try entering them into your regression model?

Note: The pipeline is a draft, and you might need to adapt it later on.